

is pushed too high or too low compared to the underlying securities these market participants jump in and perform a statistical arbitrage type function. If these participants can acquire the underlying stocks at a lower price than the ETF they will buy the shares and sell the ETF. If the ETF price is lower (due to selling pressure) than the price of the underlying securities these participants will buy the ETF and sell the stock. In theory, they will have a net zero risk position and will profit from the difference in prices. Since the acquired position is hedged—they have the exact same long and short exposure—they can trade out of both positions passively over time without incurring high trading costs and lock in a profit. Additionally, ETFs can often increase or decrease the number of its outstanding shares through creation and/or redemption. This differs from company stock that has a constant number of shares outstanding. In these cases, portfolio managers can purchase the underlying ETFs in the market if there are sufficient sellers, and if not, they can purchase the underlying equity shares and create the ETF. Alternatively, portfolio managers could sell the ETFs in the market if there are sufficient buyers, and if not, managers could redeem the shares and sell them in the market.

There is often a fee corresponding to the creation and redemption process but in these cases the fee would still be less than the incremental market impact cost for attracting necessary counterparties. The last point worth mentioning here is that ETFs do not suffer the same information content as stocks since ETF transactions are more likely believed to be due to a macro event rather than any specific company event. If there is a large buyer or seller of a stock it is more often believed to be due to company specific information such as a mispricing, undervaluation, or simply changing company fundamentals that have not yet been fully disseminated into the market.

How does this affect the ETF volumes in the market impact model? Since ETFs have corresponding “shadow” liquidity, analysts will often use a higher volume estimate than is reflected in the data. Some analysts may apply an adjustment factor, e.g., 1.5, and some analysts may attempt to measure total potential volume that could be used to create and redeem shares from the underlying stock volume data and corresponding weights in the ETF.

How about ETF market impact price sensitivity? The price sensitivity expression in the ETF market impact model a_1 is often higher than the stock model price sensitivity parameter. This is because the ETF volatility is lower due to diversification (e.g., market risk only) whereas stock